

Project: Online Silent Auction Management System

The Taylor Elementary School PTA would like a web-based system to manage their silent auction. The silent auction process consists of three phases: (1) soliciting and gathering donations, (2) organizing donations into lots and presenting them on a web site, and (3) recording bid information and contacting the winners.

I can show you documents that the silent auction committee used for the 2014 auction; however, I cannot let you have copies of some documents because they contain personal and/or proprietary information.

As a development team, your responsibility is to develop a web based system that supports all three parts of the process. Each person on your team will be responsible for supporting a different part of the process; however, your system should have the same look and feel overall and all of the parts must share the same database. If your team has two members, you can limit your system to supporting parts 1 and 2. The specific details of each part of the process are below.

Part 1: Soliciting and Gathering Donations

During this part of the process, members of the silent auction committee send out letters and/or personally contact businesses and individuals, asking for them to donate items or services. When donations are received, they need to keep track of who donated them and send a receipt to each donor for income tax purposes.

The following user requirements need to be met for this part of the system:

1. Add/modify/delete donors in the database. Prevent deletion of a donor who has items donated.
2. Create a letter asking for donations from a selected donor.
3. When donations from a person or business are received, the committee must be able to add/modify/delete the items donated.
4. Create a tax receipt that lists the individual items/services donated and their retail value by that donor with a total retail value.
5. List all of the donors that haven't received a tax receipt yet.

Suggested enhancements for this part of the system:

1. Create PDFs for the letters.
2. Implement a way to track whether a request has been sent to a donor, and be able to print letters for several donors at once.

Part 2: Organizing Donations and Presenting on Web Site

Several weeks before the actual auction begins, the members of the silent auction committee look over the donations and assign them to a lot or group them into multi-item lots. Items are grouped into a lot when they are similar and would take up less space and/or would be sold for more as a lot than individually. Each lot is photographed for inclusion on a web site.

Once the items are grouped into lots, the lots are grouped into categories. Each category represents a certain theme. For example, the category “On the town” might be created for gift certificates for restaurants, tickets to movies, and tickets to plays in the city of Norfolk. Another category might be “out of town adventures,” and consist of overnight stays in neighboring towns, tickets to amusement parks, etc. Once assigned to lots, the items should appear on a browsable web site listed by categories or be searchable by words in the description of the items. A detailed page for each lot should be available that lists the individual item(s) in the lot and shows a picture of the lot.

In preparation for auction day, the committee prints out a bidding sheet for each item that shows the Lot #, Item Description, Donated by, Retail Value, Starting Bid, and Bid Increment. The bidding sheet also has a table where bidders can write their bidder number and the amount of their bid.

The following user requirements need to be met for this part of the system:

1. Add/modify/delete lots in the database. Prevent deletion of a lot that has items assigned.
2. Assign individual items to a lot.
3. Add/modify/delete categories. Prevent deletion of a category that has lots assigned.
4. Assign individual lots to a category.
5. Create a user-facing web site that enables users to browse the categories and see lists of the lots up for auction.
6. Create a bidding sheet for each lot that shows the individual items in the lot and their retail value with a total retail value.

Suggested enhancements for this part of the system:

1. Create PDFs for the bidding sheet.
2. Enable the linking of a photograph to a lot.
3. Enable the secure upload of a photograph of a lot (must at least check file extension and mime-type).
4. Enable the searching of lots and categories using keywords.

Part 3: Recording Bid Information and Contacting Winners

In preparation for auction day, the silent auction committee prints out a set of 150 bidder forms. These are prefilled with a random, three-digit bidder number, and

have space for bidders to write their Name, Address, Cell Number, Home Number, and Email address. Bidders must first fill out this form before they can bid on any items. The bidder number appears in two places on the form – part of the form containing just the number is separated and given to the bidder for use during the auction.

After the auction is over, the high bidder on each item needs to be notified that they won, the money needs to be collected, and the items in the lot need to be delivered to them.

The following user requirements need to be met for this part of the system:

1. Add/modify/delete bidders in the database. Prevent deletion of a bidder who has won a lot.
2. Record the winning bidder and winning bid of a lot.
3. Record whether a lot has been delivered.
4. Record whether a bidder has paid for their lots.
5. Create a report that shows lots that have not been delivered with the bidder's name and their contact information.
6. Create a report that lists bidders with their contact information, showing what they owe or have paid (i.e., if they have paid, indicate that).

Suggested enhancements for this part of the system:

1. Create printable bidder forms.
2. Create a bill for a selected bidder that lists the individual lots they won, their retail value and the total amount due from that bidder.
3. Using a user-facing web site, enable bidders to bid on lots using a randomly assigned BidderID. Enable the administrators to end the auction and record the highest bid as the winning bidder for each lot.